

Ian Gao

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EDUCATION

The University of North Carolina at Chapel Hill | Chapel Hill, NC

B.S. in Computer Science | Minor in Statistics and Analytics

Expected Graduation, May 2027

Relevant Courses: Data Structures and Algorithms; Discrete Mathematics; Linear Algebra; Calculus I, II, and III; Probability and Statistics, Systems Fundamentals, Computer Organization, Models of Languages and Computation, Algorithms and Analysis, Introduction to Economics

AWARDS & ACCOMPLISHMENTS

Publication – "[Integrating Online Learning and Connectivity Maintenance for Communication-Aware Multi-Robot Coordination](#)" Co-authored and published research paper. Presented at IROS 2024 Abu Dhabi.

UNCC Undergraduate Research Presentation Award – Award for Innovative research in robotics. #1 out of 500 submissions.

Dean's List Fall 2023, Dean's List Spring 2024

EXPERIENCE

UNC Charlotte | Research Assistant | Charlotte, NC

August 2023 – December 2024

- Programmed and debugged robot navigation and communication systems using Python, focusing on real-time data processing and system reliability.
- Collaborated in a dynamic research team to develop and implement novel algorithms for multi-robot coordination, enhancing communication and operational efficiency.
- Co-authored an academic paper on advanced robotics and control systems.

PROJECTS

Communication-Aware Multi-Robot Coordination | Research Publication

May 2024

- Worked with a Professor and PhD student to develop algorithms, simulate multi-robot communication systems, and process real-world data sets to enhance connectivity and coordination in networked robotic systems.
- Implemented Data-driven Connectivity Barrier Certificates using Gaussian Processes and Control Barrier Functions.
- Awarded the Presentation Award by UNC Charlotte's Office of Undergraduate Research.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, MatLab, HTML, Javascript, CSS

Developer Tools: JUnit, Docker, Github, VS Code, Apache Commons, OpenAI API, Node.js